Safety systems · France

On the **LGV Méditerranée** (Avignon TGV included) there are no lineside signals: **TVM 430** shows the speed for each block in the cab, and the track carries only block marker boards. On the classic line into **Marseille Saint-Charles** the lineside **signaux** apply, with **KVB** checking speeds and stops and the **crocodile** repeating warning signals in the cab. The **VACMA** checks that you are awake. Speeds are in km/h. Quadrant controls are from TheJAG's TGV Duplex profile.

Safety systems

TVM 430
Transmission Voie-Machine

Cab signalling through the rails. White on green = line speed; black on a **white diamond** = reach this speed by the next marker; white on **black** = applies now; **flashing** = a further reduction follows. Overspeed brakes the train.

WHERE LGV Méditerranée

QUADRANT Left aux square = KVB, crocodile and TVM to Normal

KVB
Contrôle de Vitesse par Balises

Beacons check the speed to stop signals and limits against a braking curve: an alert 5 km/h over, then the brake. Display codes: **00 / 000** stop ahead, **P / L** limits, **FU** it has braked.

WHERE In and around Marseille

QUADRANT Left aux square = KVB, crocodile and TVM to Normal

Crocodile
Répétition des signaux (RS)

A contact ramp at warning signals. Closed-signal warning: the alert lamp flashes (with a bip); press **Acknowledge**; it stays lit until you pass the signal. Open signal: a tone only. Real world: 5 s, then the brake.

WHERE Classic line into Marseille

QUADRANT **Gear lever** = Acknowledge; left aux square = KVB, crocodile and TVM to Normal

VACMA
Veille Automatique à Contrôle de Maintien d'Appui

Real: hold the pedal; release and press again when the buzzer asks. **In TSW**: above 8 km/h a buzzer every 60 s; press Acknowledge within 5 s or the emergency brake applies.

WHERE Every SNCF train

QUADRANT **Gear lever** = Acknowledge; right aux square = VACMA to Normal

Lineside signals (cible)

Classic line only. Flashing (rays) changes the meaning: see each card. The small œilleton bottom left is lit with a sémaphore on a panel with an Nf plate.



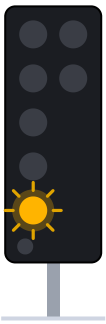
Voie libre
Clear: line speed.

DRIVE Drive on.



Feu vert clignotant
Trains above 160: be at 160 at the next signal.

DRIVE Brake to 160; crocodile tone only.



Feu jaune clignotant
Next signal is an avertissement at a short distance from its stop signal.

DRIVE Acknowledge; brake so you can stop in that short distance.



Avertissement
Be able to stop at the next stop signal.

DRIVE Acknowledge the crocodile; brake to stop.



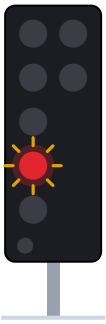
Sémaphore
Stop (block signal, œilleton lit).

DRIVE Stop; then on sight at max 30 if allowed.



Carré
Absolute stop (interlocking); œilleton dark.

DRIVE Stop; pass only on an order.



Feu rouge clignotant
Pass without stopping, on sight to the end of the block.

DRIVE Max 15 km/h, ready to stop.



Ralentissement 30
Points at 30 at the next signal.

DRIVE Acknowledge; be at 30 at the next signal.



Ralentissement 60
Points at 60 at the next signal.

DRIVE Acknowledge; be at 60 at the next signal.



Rappel 30 / 60
Steady 30, flashing 60 over the points at this signal.

DRIVE Hold it until the rear has cleared the points.

Sources: Dovetail Games: LGV Méditerranée driver's manual · SNCF S1A (IN 1482) Signalisation au sol · SNCF S1C (IN 1493) Répétition des signaux, contrôles de vitesse · fr.wikipedia: Transmission voie-machine, Contrôle de vitesse par balises, Crocodile, VACMA

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Safety systems · France

TVM 430 cab display

Typical TVM 430 steps on a 320 line: 320, 300, 270, 230, 170, 80, 000; other values appear in TSW.



Line speed

White on green: line speed here and in the next block.

DRIVE Drive at the line speed.



Announcement

Black on a white diamond: the next block is limited to this.

DRIVE Brake to reach it by the next marker.



Execution

White on black: this speed applies now.

DRIVE Hold at or below it.



000

Stop ahead: be able to stop before the next marker.

DRIVE Stop at an Nf marker; an F marker may be passed.



Red squares

Full stop within range / rouge cabine.

DRIVE Stop; on sight at max 30 if allowed.

Boards



Repère TVM

Block marker; the triangle points at its track. Nf = do not pass at 000.

DRIVE Reach the displayed speed by the marker.



Entrée CAB

Start of TVM cab signalling (an advance board comes first).

DRIVE Follow the cab display from here.



Sortie CAB

End of TVM: lineside signals from here.

DRIVE Watch the signals; KVB and crocodile take over.



TIV à distance

A lower limit ahead (lozenge = with crocodile).

DRIVE Acknowledge; brake to reach it by the Z board.



Z

Start of the speed limit.

DRIVE Be at the limit here.



R

End of the speed limit.

DRIVE Accelerate once the rear has passed.

How to drive under TVM, KVB, the crocodile and the VACMA

TVM 430 on the LGV

1. Drive to the speed on the cab display.
2. **White diamond**: brake now so you reach it by the next block marker.
3. **White on black**: hold at or below it in this block.
4. **Flashing**: the block after the next is slower still.
5. **000**: be able to stop at the next marker (Nf: stop there).

Crocodile and KVB near Marseille

1. At a closed-signal warning the alert lamp flashes: press **Acknowledge** (gear lever).
2. Brake for the next signal; KVB checks your curve (00 / 000 on its display).
3. KVB overspeed: alert, then the emergency brake (FU). Stop, Acknowledge, reset the controls, drive on on sight.

VACMA

1. Keep the VACMA at Normal (right aux square).
2. Above 8 km/h the buzzer sounds every 60 s: press **Acknowledge** (gear lever) within 5 s.
3. Missed: emergency brake. Wait for a complete stop, Acknowledge, release and drive on.

Before departure

1. KVB / crocodile / TVM and VACMA to Normal (aux squares lit): the train loads with them isolated.
2. Check the signal or the TVM display, and the doors.
3. Frein direct to Release, handle into power.

At a signal at stop

- Sémaphore (œilleton lit): stop; then on sight at max 30 km/h where allowed.
- Carré: stop; pass only on an order from the agent-circulation.
- Feu rouge clignotant: no stop, on sight at max 15 km/h.

Sources: Dovetail Games: LGV Méditerranée driver's manual · SNCF S1A (IN 1482) Signalisation au sol · SNCF S1C (IN 1493) Répétition des signaux, contrôles de vitesse · fr.wikipedia: Transmission voie-machine, Contrôle de vitesse par balises, Crocodile, VACMA

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